MILITARY SPECIFICATION

REGULATOR, 650 AMP, SOLID STATE

This specification approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state voltage regulators for use primarily on military combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
MILITARY

MIL-P-15024	Plates, Tags and Bands for Identification of Equipment.
MIL-G-62186	Generator, Alternating Current (650 Ampere Rectified), 28 Volts Direct Current (Oil Cooled).

STANDARDS
FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	Dissimilar Metals.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

11655469	- Generator AC/DC (Oil Cooled).
12354334	- Regulator, Solid State.

(Copies of specifications, standards and drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish regulators which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent regulators delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein, in applicable specifications, and on applicable drawings. Materials not specifically designated shall be of good commercial quality. Materials used shall be suitable for use in the regulator operating over an ambient temperature range of -65 to 225 degrees Fahrenheit (°F) without any change in physical or dimensional properties that would result in operation of the regulator falling outside of the limits specified (see 4.6.1).

3.2.1 Dissimilar Metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces. Dissimilar metals are defined in MIL-STD-889 (see 4.6.1).

3.2.2 External Surfaces. All external surfaces shalL ‘be corrosion resistant (see 4.6.1).

3.2.3 Fungicidal Materials. All non-metallic materials shall be inherently fungicidal or treated to resist fungus growth except when used for components of potted compounds or otherwise moisture sealed assemblies (see 4.6.1).

3.2.4 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.6.1 and 6.3.1).

3.3 Design and construction. Regulators acquired in accordance with this specification shall conform to Drawing 12354334 and specifications and standards to the extent specified herein (see 4.6.1, 4.6.2 and 6.2).

3.3.1 Interchangeability. All regulators shall be interchangeable with respect to installation and performance. The voltage regulator after calibration at time of manufacture or overhaul shall not require any adjustment at time of installation. To provide for interchangeability, the assembly fabricated in accordance with this specification shall have external dimensions conforming to the envelope outline of Drawing 12354334. Electrical connectors, mounting dimensions, and location shall conform to the detail requirements of Drawing 12354334. Electrical contact designation shall conform to requirements of 12354334 (see 4.6.2).

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3.3.2 Electrical Interface. Electrical connector pin designations to the utilizing vehicle shall be provided as follows (see 4.6.2).

Connector pin function	Connector pin designations
Regulation Sensing Input	A
Current Limit Signal Input	B
Ground	C
Generator Alternator AC Signal Input	D
Generator Alternator Field Control Output	E
Accessory Relay Signal Output	F
Overvoltage Test Circuit Input	G

3.3.3 Standard hardware. Military standard parts shall be used wherever possible. Screw threads shall conform to FED-STD-H28 (see 4.6.2).

3.3.4 Generator-Alternator compatibility. The regulator shall function as specified with generator-alternators conforming to MIL-G-62186 and Drawing 11655469 (see 4.6.1).

3.3.5 Installation compatibility. The regulator shall fit and perform as specified herein when a trial installation with the generator-alternator (see 3.3.4) is specified on the M60A3 tank (see 4.6.1 and 6.2).

3.3.6 Weight. The regulator shall not exceed a weight of 6 pounds (see 4.6.2).

3.4 Performance.

3.4.1 Voltage regulation. The regulator shall establish and maintain output voltage in accordance with figure 1 throughout the generator-alternator speed range of 2400 to 8000 revolutions per minute (rpm) (see 4.7.1).

3.4.2 Ripple voltage. Ripple voltage shall not exceed 4 volts (V) peak to peak with output voltage at 28 V direct current (dc) throughout the generator-alternator speed range of 2400 to 8000 rpm (see 4.7.2).

3.4.3 Overvoltage. The overvoltage portion of the regulator shall deactivate the field circuit when the system voltage (E_s) reaches 33.0 ± 1.0 V. Deactivation and return to battery voltage shall be accomplished in .07 seconds. Reset shall be manual and shall be located on the regulator (see 4.7.3).

3.4.4 False trips. False tripping of the regulator shall not be caused by buildup, load switching, and input speed acceleration or deceleration (see 4.7.4).

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3.4.5 Current limit. The current limiting shall be incorporated into the regulator assembly which will limit the generator-alternator output current controlled by the regulator in accordance with figure 2. Stable current limiting shall be established within 0.05 seconds. System voltage undershoot shall not be below 18 V (see 4.7.5).

3.4.6 Accessory relay control. The accessory relay shall close when the generator-alternator speed is between 900 and 1100 rpm. This relay must open before the generator-alternator speed reaches 800 rpm during shutdown (see 4.7.6).

3.4.7 Fault protection. Circuitry shall be included in the regulator to deactivate the field circuit in the event of wiring malfunctions. The circuitry is not required to operate if a malfuntion in any of the trip types of table I does not create a condition which would damage the regulator or associated electrical power sources. Maximum voltage (V), amperes (A) and trip time limits are as follows for the three trip types (see figure 3) (see 4.7.7):

TABLE I. Trip types.

Trip type	Generator-Alternator rpm	E_A V	E_s V	I_f A	Time seconds
Regulator Sensing Input Lead	2000	35.0	35.0	3	0.06
	8000	35.0	35.0	4	1.00
Alternator Positive Output Lead	2000	40.0	30.0	4	1.50
	8000	115.0	30.35	2	1.70
Alternator Negative Output Lead	2000	35.0	30.5	5	2.0
	8000	55.0	30.5	1	0.4

3.4.8 Dielectric strength. The regulator circuit insulation to ground shall not permit a leakage current greater than 5.0 milliamperes (ma) with 500 V alternating current (AC) applied between all receptacle pins at the same time and the case with the case ground lead disconnected (see 4.7.8).

3.4.9 Short circuit protection. Regulator circuitry shall be provided to prevent damage to internal components in the event of a short circuit from pin “E” or “F” to ground during normal regulator operation (see 4.7.9).

3.4.10 Endurance. The regulator shall meet the following requirements before and after 3000 hours (125 cycles) of operation conforming to the time schedule and conditions set forth in table II. During the endurance test, the regulator shall require no servicing adjustment, or replacement of parts. After endurance test the regulator shall meet the following requirements (see 4.7.10).

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3.4.1 Voltage regulation.
3.4.2 Ripple voltage.
3.4.3 Overvoltage.
3.4.5 Current limit.
3.4.6 Accessory relay control.
3.4.7 Fault protection.
3.4.8 Dielectric strength.
3.4.12 Case Grounding.
3.5.5 Waterproofness.
3.4.9 Short circuit protection.

TABLE II. Regulator endurance time schedule and conditions.

1/ Time hours	2/ RPM	3/ System volts	Alternator resistive load			Load cycles minutes	Cyclic load seconds	
			Constant current	Cyclic current	Total current		On	Off
1.5	2400	26.5 ± 0.50	320	330	650	90	3	57
7.5	5120	26.5 ± 0.50	320	330	650	450	3	57
14.5	3520	26.5 ± 0.50	320	330	650	870	3	57
.5	8000	26.5 ± 0.50	650	0	650	30	-	—

1/ 24.0 hours per cycle.
2/ Alternator.
3/ System V in Table II apply at +225°F (see 3.4.7). See figure 4 for voltage values at lower temperatures.

3.4.11 Electromagnetic interference. The regulator shall not exceed the broad band conducted and radiation emission limits of figures 6 and 7 (see 4.7.11).

3.4.12 Case rounding. The regulator shall be electrically connected to the ground circuitry (see 4.7.12).

3.4.13 Reverse polarity protection. The regulator circuit shall not be damaged in the event reverse polarity is applied to the input power connector (see 4.7.13).

3.5 Environmental conditions. The regulator shall meet the requirements of 3.4 after being subjected to the conditions specified herein.

3.5.1 Shock resistance. When mounted as in intended use, the regulator shall withstand 40 gravities (g) peak saw tooth shock pulses for an 18 ± 3 millisecond (ms) duration in each of the major mutually perpendicular axes (see 4.8.1).

3.5.2 Vibration resistance. The assembly, when mounted as in intended use, shall withstand an input vibration maximum level of 4 g at a sweep frequency range of 5 to 500 Hertz (Hz) for a period of 100 minutes in each of three mutually perpendicular axes (see 4.8.2).

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3.5.3 Corrosion resistance. The regulator shall meet performance requirements of 3.4 after exposure to 192 hours of salt fog (see 4.8.3).

3.5.4 Humidity resistance. The regulator shall meet performance requirements of 3.4 after exposure to humidity (see 4.8.4).

3.5.5 Water proofness. The assembly shall withstand 10 pounds per square inch gage (psig) internal pressure and 10 psig internal vacuum without leakage and while operating in salt water (see 4.8.5).

3.5.6 Fungus resistance. The regulator shall meet the requirements of 3.4 after 28 days exposure to fungal incubation (see 4.8.6).

3.5.7 Extreme temperature resistance. The regulator shall meet the performance requirements of 3.4 during and after operation at extreme ambient temperatures. The range of regulated voltage over the total operational temperature range shall conform to figure 4 (see 4.8.7).

a. High temperature, + 225°F.
b. Low temperature, - 65°F.

3.6 Identification and marking (see 4.6.2).

3.6.1 Marking. The regulator shall be marked as specified on the applicable drawing and in conformance to MIL-STD-130.

3.6.2 Nameplate. A nameplate conforming to MIL-P-15024, Type A, shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the nameplate shall include the following as a minimum (see 4.6.2).

Regulator, solid state
28 Volts DC
Army part number
National stock number
Manufacturer's identification (CAGE)
Manufacturer's serial number
Contract number
Date of manufacture
US

3.7 Workmanship. The regulator shall be free of rust, burrs and scratches. Solder connections shall conform to requirement 5 of MIL-STD-454. All solder cleaning agents and flux shall be chemically neutralized and removed after soldering. Manufacturing methods shall not cause degradation of the inherent reliability and durability of the regulator (see 4.6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions.

4.3.1 Atmospheric limits. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 ± 15°F
b. Barometric pressure 28.5 + 2 inches of mercury
- 3
c. Relative humidity 50 ± 30%

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4.3.2 Laboratory test conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government with the following procedures:

a. Oscillographic type records shall have a frequency response from 0 to 2000 Hertz which is flat within ± 5.0%.
b. Prior to each performance test, the regulator must be operated for 15 minutes at 3000 rpm and at 650 A alternator current for temperature stabilization.
c. Electrical switches SW8, SW9 and SW13 (see figure 3) shall be closed at all times unlesS otherwise specified and SW3 is always open unless otherwise specified. SW1 shall be the first closed when energizing the regulator followed by SW2. SW4 shall be closed whenever resistive load is required during these tests. When de-energizing the system, SW1 shall always be the last switch opened unless otherwise specified. SW1 shall always-be the last switch opened unless otherwise specified. SW1 shall be opened and closed only with the rotary power source at rest.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four regulator assemblies produced under the production contract for first article inspection. First article testing shall be conducted by the contractor under Government surveillance, and shall consist of defects specifed in 4.6.2 and tests specifed in table III. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply regulator assemblies that are fully representative of those inspected as a first article sample.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two regulators. Preproduction inspection shall consist of inspection as specified in table III.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two regulators from the first ten produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table III.

4.4.3 First article inspection failure. Test item deficiencies found during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all item shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

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4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the regulator assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article		Quality conformance		Control
			PPI	IPT	Examination	100% Tests	
Materials and construction	3.2 thru 3.2.4, 3.3.4 and 3.3.5	4.6.1	X	X			1/300
Defects	3.3 thru 3.3.3, 3.3.6, 3.6 and 3.7	4.6.2	X	X	X		
Comprehensive performance:							
Voltage regulation	3.4.1	4.7.1	X	X			
Ripple voltage	3.4.2	4.7.2	X	X			
Current limit	3.4.5	4.7.5	X	X			
Current limit transient	3.4.5	4.7.5	X	X			
Current limit regulation	3.4.5	4.7.5	X	X			1/300
Accessory relay	3.4.6	4.7.6	X	X			
Fault protection	3.4.7	4.7.7	X	X			
Short circuit protection	3.4.9	4.7.9	X	X			
Overvoltage	3.4.3	4.7.3	X	X			
Dielectric strength	3.4.8	4.7.8	X	X			
Waterproofness	3.5.5	4.8.5	X	X			1/50

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TABLE III. Classification of inspections - Continued.

Title	Requirement	Inspection	First article		Quality conformance		Control
			PPI	IPT	Examination	100% Tests	
Abbreviated performance:							
Voltage regulation	3.4.1	4.7.1.1	X	X		X	
Ripple voltage	3.4.2	4.7.2.1	X	X		X	
Current limit	3.4.5	4.7.5.1	X	X		X	
Acceptance relay	3.4.6	4.7.6	X	X		X	
Fault protection	3.4.7	4.7.7	X	X		X	
Short circuit protection	3.4.9	4.7.9	X	X		X	
Overvoltage	3.4.3	4.7.3.1	X	X		X	
Dielectric strength	3.4.8	4.7.8	X	X		X	
Case grounding	3.4.12	4.7.12	X	X		X	
Reverse polarity protection	3.4.13	4.7.13	X	X		X	
Waterproofness	3.5.5	4.8.5.1	X	X		X	
Shock resistance	3.5.1	4.8.1	X	X			
Control shock resistance		4.8.1.1					1/50
Vibration resistance	3.5.2	4.8.2					
Control vibration resistance		4.8.2.1					1/50
Corrosion resistance	3.5.3	4.8.3	X	X			
Humidity	3.5.4	4.8.4	X	X			
Fungus resistance	3.5.6	4.8.6	X	X			
High temperature	3.5.7	4.8.7	X	X			1/100
Low temperature	3.5.7	4.8.7	X	X			1/100
Endurance	3.4.10	4.7.10	X	X			
Endurance control	3.4.10	4.7.10.1					1/300
Electromagnetic interference	3.4.11	4.7.11	X	X			

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4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Materials not certified (see 3.3).	Visual
102	Incorrect dimensions affecting mounting and interchangeability (see 3.3 and 3.3.1).	SIE 1/
103	Contacts other than specified (see 3.3.2).	visual
104	Non-military standard parts where military standard parts are applicable (see 3.3.3).	Visual and SIE
105	Nonconforming weight (see 3.3.6).	SIE
106	Faulty workmanship affecting performance (see 3.7).	Functional
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3 and 3.3.1).	SIE
202	Improper marking and nameplates (see 3.6.1 and 3.6.2).	visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test (100%). Each regulator shall be subjected to the quality conformance tests specified in table III.

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4.5.4 Control inspection. Unless otherwise specified, inspections shall be conducted by the supplier for maintaining control of those requirements whose length, expense, or destructiveness prohibit their inclusion under lot-by-lot inspection. Samples shall be selected and subjected to the inspections with the frequencies designated in table III. These inspections need not be performed on the first control inspection quantity when initial production inspection has been satisfactorily accomplished.

4.5.4.1 Inspection frequency of one in fifty (1/50). Inspections as indicated in table III shall be performed on one sample for each 50 produced or on one sample every two months, whichever occurs first.

4.5.4.2 Inspection frequencies other than one in fifty. Inspections shall be performed on one sample at the interval specified in table III. The sample may be selected from either the first ten units produced for the internal defined, or the last ten units produced from the preceding interval.

4.5.5 Failure. Failure of any regulator assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.2 through 3.3.1 and 3.6 through 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7 Performance.

4.7.1 Voltage regulation. To determine conformance to 3.4.1 install the regulator in a circuit as shown in figure 3 and operate the regulator. With SW1 and SW2 closed, take oscillographs of regulated voltage (Es) while resistive loads are applied and removed. These resistive loads shall cause a current flow of 650 A at 2400, 4000 and 8000 rpm. E_s shall be within the curves shown on figure 3.

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4.7.1.1 Abbreviated voltage regulation. With the stabilized regulator installed in a circuit as in 4.7.1 the output voltage shall be observed during the following procedures. The resistive load shall be applied and removed suddenly as indicated.

a. Operate generator-alternator at 2400 rpm with resistance load at zero (battery only).
b. With generator-alternator at 2400 rpm, add resistance load to establish a total current flow of 100 A.
c. With generator-alternator speed at 2400 rpm, increase resistive load to establish a current flow of 650 A.
d. Increase generator-alternator speed to 4000 rpm, and remove the resistance load.
e. With generator-alternator speed at 4000 rpm add resistance load to establish a total current flow of 100 A.
f. With generator-alternator at 4000 rpm, increase resistance load to establish a current flow of 650 A.
g. Increase generator-alternator speed to 6000 rpm and remove the resistance load.
h. With generator-alternator at 6000 rpm, add resistance load to establish a total current flow of 100 A.
i. With generator-alternator speed at 6000 rpm, increase resistive load to establish a current flow of 650 A.
j. Increase generator-alternator speed to 8000 rpm and remove the resistance load.
k. With generator-altenator at 8000 rpm, add resistance load to establish a total current flow of 100 A.
l. With generator-alternator speed at 8000 rpm, increaae resistive load to establish a current flow to 650 A.

4.7.2 Ripple voltage. To determine conformance to 3.4.2, the regulator shall be connected in a circuit as shown in figure 2 and operated at the rate and load to produce a current flow as indicated below. Ripple voltage shall be measured at point E_s and recorded from oscillograph readings for these conditions for peak to peak and for +peak to -peak.

rpm	Current, A (I_A)
3000	50
3000	650
8000	50
8000	650

4.7.2.1 Abbreviated ripple voltage. The regulator shall be tested as in 4.7.2 except that the test point will be 8000 rpm at a load producing a current flow of 650 amperes (I_A).

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4.7.3 Overvoltage. To determine conformance to 3.4.3, the regulator shall be connected in a circuit as shown in figure 3 and operated as follows. With SW1 and SW2 closed, adjust alternator speed to 5000 rpm. Record system voltage (E_S) on an oscillograph as SW3 is closed. Record E_S where trip occurs. After recording data open SW3 and reset overvoltage switch on regulator. Read the trip voltage and time required for a return to battery voltage from the oscillograph.

4.7.3.1 Abbreviated overvoltage. With the regulator connected as shown in figure 5, increase power supply voltage until overvoltage circuit breaker trips. Record DC voltmeter reading only at the overvoltage switch trip point. Disconnect DC power and reset regulator overvoltage switch.

4.7.4 False trips. To determine conformance to 3.4.4, operate the regulator with no external alternator load. The overvoltage switch must trip after each step when steps a and b are performed in sequence.

a. Open SW8.
b. Close SW8, reset overvoltage switch and open SW9.

Then, close SW9 and reset the overvoltage switch. Verify that the regulator operates without tripping.

4.7.5 Current limits. To determine conformance to 3.4.5, install the regulator in a circuit such as shown in figure 2 and perform the test below.

a. Current Limit, Transient. Open switch SW13 and operate the regulator for 15 minutes with a 500 ampere alternator current flow at 5000 rpm. Disconnect all load from the alternator except batteries. Supply a resistive load to cause a current flow of 1000 A to the alternator. Alternator current (I_A) and system voltage (E_S) shall be recorded on an oscillograph tape and read to determine conformance with the recovery and undershoot requirements of 3.4.5.

b. Current Limit, Regulation. Apply a resistive load to produce a current flow of 50 A to the alternator at 2400 rpm. Increase the loading in 50 A increments until the system voltage is reduced to less than 24 V. Measure the system voltage and current to determine conformance to figure 2. Repeat at 5000 rpm alternator speed.

4.7.5.1 Abbreviated current limit. With the regulator installed as in 4.7.5, and operationally stabilized and with the alternator at 2400 rpm, apply a resistive load to produce a current flow of 400 A to the alternator, record output at E_S on an oscillograph tape. Open switch SW13 and record output at E again. The Es readings must be within 0.2 V of each other. Increase the load and record current at I_A and E_S. Over the speed and temperature range, the current limit shall conform to figure 2.

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4.7.6 Accessory relay control. To determine conformance to 3.4.6, test the regulator while it is installed in a circuit as is shown in figure 3. With SW1 and SW2 closed, slowly increase alternator speed from zero rpm until the accessory relay closes (light at L1 comes on). Now slowly decrease the alternator speed until the accessory relay opens (light L1 goes out). Record alternator speeds at which relay closes and opens.

4.7.7 Fault protection circuitry. Install the regulator in a circuit such as is shown in figure 3. Perform the following tests to determine conformance to 3.4.7. Record E_s, E_A and If on an oscillograph for each of the faults described below:

a. Regulator sensing. input lead.

1. Operate the regulator and alternator at 2000 rpm with no external alternator load with SW10 closed. Set variable resistance R10 at zero ohms and open SW10. Slowly increase R10 until the protective circuit trips. Record E_A, E_s and I_f just before the trip occurs. Leave R10 at the final setting. Close SW10 and reset the regulator overvoltage switch.

2. Operating at 2000 rpm, open SW1O while recording E_A, E_s and I_f on an oscillograph. After the circuit trips, close SW10 and reset regulator overvoltage switch.

3. Repeat a1 above at 8000 rpm.

4. Repeat a2 above at 8000 rpm.

b. Alternator positive output cable.

1. Operate the regulator and alternator at 2000 rpm, with no external alternator load with SW11 closed. Set R11 at zero ohms and open SW11. Slowly increase R11 until the protective circuit trips.
before the trip occurs. Leave R11 at the final setting. Close SW11 and reset overvoltage switch.

2. Operating at 2000 rpm, open SW11 while recording E_A, E_S and If on an oscillograph. After the circuit trips, close SW11 and reset the overvoltage switch.

3. Repeat b1 above at 8000 rpm.

4. Repeat b2 above at 8000 rpm.

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c. Alternator negative output cable.

1. Operate the regulator and alternator at 2000 rpm, with no external alternator load with SW12 closed. Set R12 at zero ohms and open SW12. Slowly increase R12 until the protective circuit trips. Record E_A, E_s and I_f just before the trip occurs. Leave R12 at the final setting. Close SW12 and reset overvoltage switch.

2. Operating at 2000 rpm, open SW12 while recording E_A, E_s and I_f on an oscillograph. After the circuit trips, close SW12 and reset the overvoltage switch.

3. Repeat c1 above at 8000 rpm.

4. Repeat c2 above at 8000 rpm.

4.7.8 Dielectric strength. To determine conformance to 3.4.8, remove the cover and disconnect the internal circuit ground lead from the case. Connect all the pins of the regulator receptacle together. Connect a Bendix Model 60B4-1 insulation tester or its equivalent between the shorted pins and the bare metal case. Starting with the taster voltage at 0.0 V, slowly increase the voltage to 500.0 V AC for 1 minute and then decrease the tester voltage to 0 before removing the test leads. Reconnect the circuit ground lead to the case after test and replace the cover.

4.7.9 Short circuit protection. To determine conformance to 3.4.9, connect the regulator to the test configuration as shown in figure 8 and operate the regulator according to the matrix of table V.

TABLE V. Short circuit protection.

SWI	SW2	Circuit Breaker 1/	LI
Open	Open	On	On
Close	Open	Off	Off
Open	Open	Off	Off
Open	Open	On	On
Open	Close	Off	Off
Open	Open	Off	Off
Open	Open	On	On

1/ Breaker will automatically trip off, but must be reset by switching on by hand.

A test will be considered successful when the matrix is followed in the order shown with the resulting circuit breaker and lamp (L1) response. The regulator must successfully pass test 4.7.1 before and after this short circuit test is performed to insure no circuit has been damaged.

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4.7.10 Endurance. To determine conformance to 3.4.10, the regulator shall be subjected to 3000 hours of operation per table II at an ambient temperature of 225 ± 5°F. The following listed data shall be recorded at the beginning and end of each speed change. After endurance testing the regulator shall be tested per 4.7.1. Record the following data:

a. I_A alternator current.
b. Alternator rpm.
c. If field current.
d. E_f field voltage.

4.7.10.1 Control endurance. To determine conformance to 3.4.10 for control inspections, the selected regulator shall be installed in a circuit as shown in figure 3, and operated for 4 sequential cycles per table II for a total of 96 hours. During this test the regulator shall require no servicing adjustments or replacements of parts. The regulator shall than be tested per 4.7.1.1.

4.7.11 Electromagnetic (EMI) interference. To determine conformance to 3.4.11, the regulator shall be interconnected with alternator, P/N 11655469 and subjected to EMI tests for radiated and conducted broadband emissions in accordance with MIL-STD-462, Method CE04 and RE02 see 6.1).

4.7.12 Case grounding. To determine conformance to 3.4.12, verify continuity between the regulator case and pin C of the connector while the regulator is disconnected from the system.

4.7.13 Reverse polarity protection. To determine conformance to 3.4.13, apply 24 V dc to pin C and ground to pin A for 1 minute with the regulator circuit breaker switched on. The regulator shall meet performance test 4.7.1 before and after applying reverse polarity voltage to insure no circuits have been damaged due to testing.

4.8 Environmental conditions.

4.8.1 Shock resistance. To determine conformance to 3.5.1, the assembly shall be mounted as in use and subjected to three sawtooth shock pulses of 40g at 18 ± 3 ms duration in each direction along three mutually perpendicular axes for a total of 18 shock pulses. The sawtooth shock pulse configuration and tolerancs limits shall be as shown for ground equipment in figure 516.3-4 516-4 of MIL-STD-810.

4.8.1.1 Control shock resistance. The regulator will be tested as in 4.8.1 except that the shock pulse amplitude shall be 30g.

4.8.2 Vibration resistance. To determine conformance to 3.5.2, the regulator shall be mounted by normal mounting provisions to a vibration exciter capable of producing and of transmitting the vibrations specified herein. The regulator shall be connected as in use and also connected to the instrumentation necessary for this test. Vibration shall be for 100 minutes in each direction along each of three perpendicular axes. Amplitude shall be

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4 gravity units. Frequency shall be 5 to 500 to 5 Hz in a 15 minute time sweep. Test level shall be in accordance with the curve of figure 10. Resonent frequencies of the regulator shall be determined and a sinusoidal resonance search shall be conducted for the four most severe resonant frequencies of the unit. The regulator shall be subjected to 16.34 minutes at each severe resonent frequency within each 100 minutes sweep test. The frequency of applied vibration shall be swept over the specified range in accordance with figure 9. The sweep time is that of an ascending plus a descending time shown on figure 9. The vibration resistance test shall be performed at an ambient temperature of -65°F and 225°F.

4.8.2.1 Control vibration resistance. For control testing, the regulator will be tested as in 4.8.2 except that dwell time at each resonance shall be 2 minutes and vibration shall be performed at room ambient temperature.

4.8.3 Corrosion resistance. To determine conformance with 3.5.3, the regulator shall be exposed to salt fog of MIL-STD-810, Method 509 for four test cycles (192 hours total). The regulator assembly shall be sealed so that internal components are not exposed.

4.8.4 Humidity resistance. To determine conformance to 3.5.4, the regulator shall be exposed to the following procedure:

a. The regulator shall be installed as in use in a chamber vented and baffled to assure an even flow of warm humid air as specified herein. No condensation shall be permitted to drip on the regulator. Relative humidity shall be measured by the dry bulb-wet bulb thermometer comparison method or approved equivalent. Air velocity in the internal test space shall not exceed 150 feet per minute. Steam or distilled, demineralized or deionized water having a pH value between 6.0 and 7.2 at 73 ± 3°F shall be used to obtain the specified humidity. No rust or corrosize contaminants will be imposed on the teat sample by the test facility. During this test no repair or replacement shall be permitted; the regulator shall be operated only when operational test data is required.

b. After having been dried for 24 hours at 129± 3°F, the regulator shall be conditioned at 73 ± 3°F and 50 ± 10% relative humidity for 24 hours. Pretest data shall be obtained by performing the tests specified in 4.7.1. Raise the internal chamber temperature to 86 ± 3°F and 94 ± 4% relative humidity. Expose the regulator to 5 continuous 48-hour cycles in accordance with figure 11. Accumulated" moisture may be removed by turning the regulator upside down or shaking. Wiping is not permitted. Observe that the regulator shall have no visible damage, corrosion or swelling as a result of exposure to humidity. Verify that the regulator will meet performance requirements by testing as specified in 4.7.1.

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4.8.5 Waterproofness. To determine conformance to 3.5.5, the regulator shall be subjected to internal conditions of 10 psig pressure. Using a 1/8" adapter fitting, connect a vacuum pump to the regulator. Temperature of the liquid and test sample shall be 73 ± 18°F. Submerge the regulator into a salt water bath (see 4.8.3) with a minimum depth of 1 inch covering the unit. Apply 10 psig of vacuum and operate in accordance with 4.7.1. After vacuum test, remove unit from water bath and thoroughly dry outside of test sample. Carefully remove the cover and inspect the regulator for water leakage. Evidence of moisture in or on the regulator interior is not acceptable. Carefully replace cover, remove the vacuum pump and connect to a dry nitrogen source. Submerge the regulator as before, apply 10 psig dry nitrogen pressure and test in accordance with 4.7.1. Escaping bubbles from the regulator interior during this test is not acceptable.

4.8.5.1 Abbreviated waterproofness. For acceptance test, the regulator shall be tested as in 4.8.5 except that operation, during and after submergence shall conform to 4.7.1.1.

4.8.6 Fungus resistance. To determine conformance to 3.5.6, the regulator shall be subjected to fungal incubation for 28 days in accordance with Method 508 of MIL-STD-810. Incubate samples under cyclic temperature and humidity conditions to include 20 hours of relative humidity at 95 ± 5% at an air temperature of 86 ± 2°F followed by 4 hours of 100% relative humidity at 77 ± 2°F.

4.8.7 Extreme temperature resistance. To determine conformance to 3.5.7, test the regulator in a chamber suitable for the development and maintenance of the specified temperatures (reference MIL-STD-810, Methods 501 and 502). The regulators shall be tested per 4.7.1 before, during and after each test specified below.

a. Low temperature.

1. Storage: After stabilization in the chamber, the regulator shall be kept at -80°F for 12 hours.
2. Operation: After storage at -80°F, the temperature shall be raised to -65°F. When stabilized at this temperature the regulator shall be tested as specified in 4.7.1.1, 4.7.2.1, 4.7.3.1, 4.7.4, 4.7.5.1, 4.7.6, 4.7.7, 4.7.9, 4.7.10.1 and 4.7.12. Verify that the voltage values conform to figure 4.

b. High temperature.

1. With the regulator installed as ready to operate, raise the internal chamber temperature to 120°F and maintain the temperature for 6 hours. Then, increase the internal chamber temperature to 160°F within a 1-hour time period. Maintain this temperature for an additional 4 hours. Then lower the internal chamber temperature to 120°F within a 1-hour period. Repeat this cycle three times for a total of 36 hours. Adj ust

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the internal chamber temperature to 225°F and maintain until the regulator is temperature stabilized. Operate and test the regulator in accordance with 4.7.1.1, 4.7.2.1, 4.7.3.1, 4.7.4, 4.7.5.1, 4.7.6, 4.7.7, 407.9, 4.7.10.1, 4.7.11 and 4.7.12. Verify that the voltage values conform to figure 4. Return and stabilize the regulator to standard ambient conditions. Operate and verify that no operational deficiencies or physical damage exists.

4.8.7.1 Extreme temperature control test. For control testing, the high and low temperature tests of 4.8.7 should be performed except that performance of the regulator during and after extreme temperature exposure shall be tested as specified in 4.7.1, through 4.7.7 and 4.7.10, through 4.7.12.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The regulator is intended to monitor the alternator to provide prescribed electrical power for engine driven military surface vehicles. The EMI requirements herein, are applicable only to the use of this item in M60 Series Tanks (see 4.7.11).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first articles samples are not required (see 3.1).
c. Other applicable specifications or standards if required (see 3.3).
d. If trial installation on an M60A3 tank is required (see 3.3.5).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.3).
h. If the number of first article samples shall be other than as specified (see 4.4).
i. If the inspection lot samples shall be other than as specified (see 4.5.1.1).
j. Selection of applicable level and packaging requirements (see 5.1).

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6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Current and voltage regulator, 650 A, solid state.
Solid state, 650 A voltage regulator.
Voltage regulator, 650 A, solid state.

6.5 Supersession data. This military specification supersedes purchase description ATPD MIL-P-62187(AT), dated 11 December 1973.

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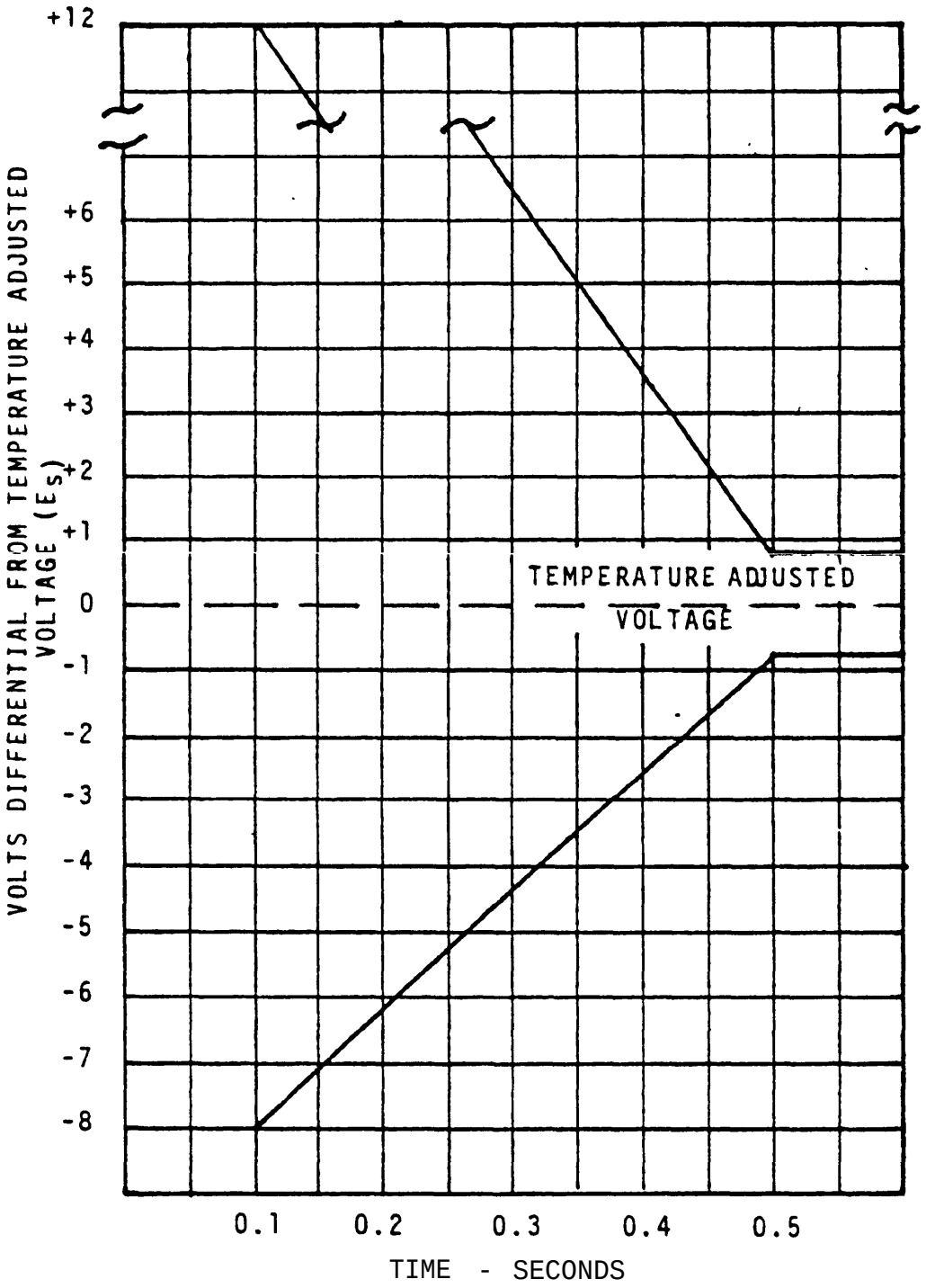


FIGURE 1. Voltage regulation

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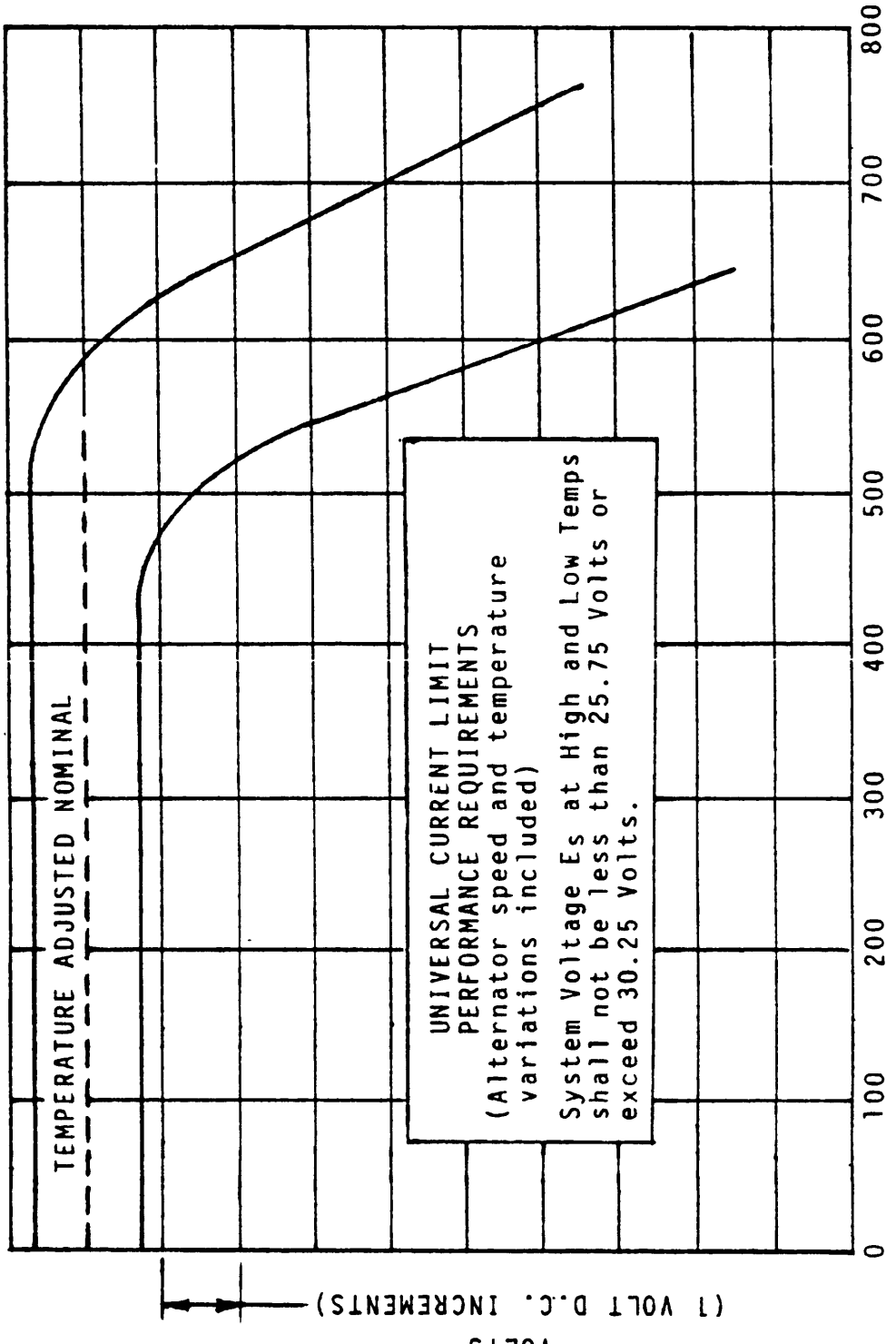


FIGURE 2. Universal current limit performance requirements.

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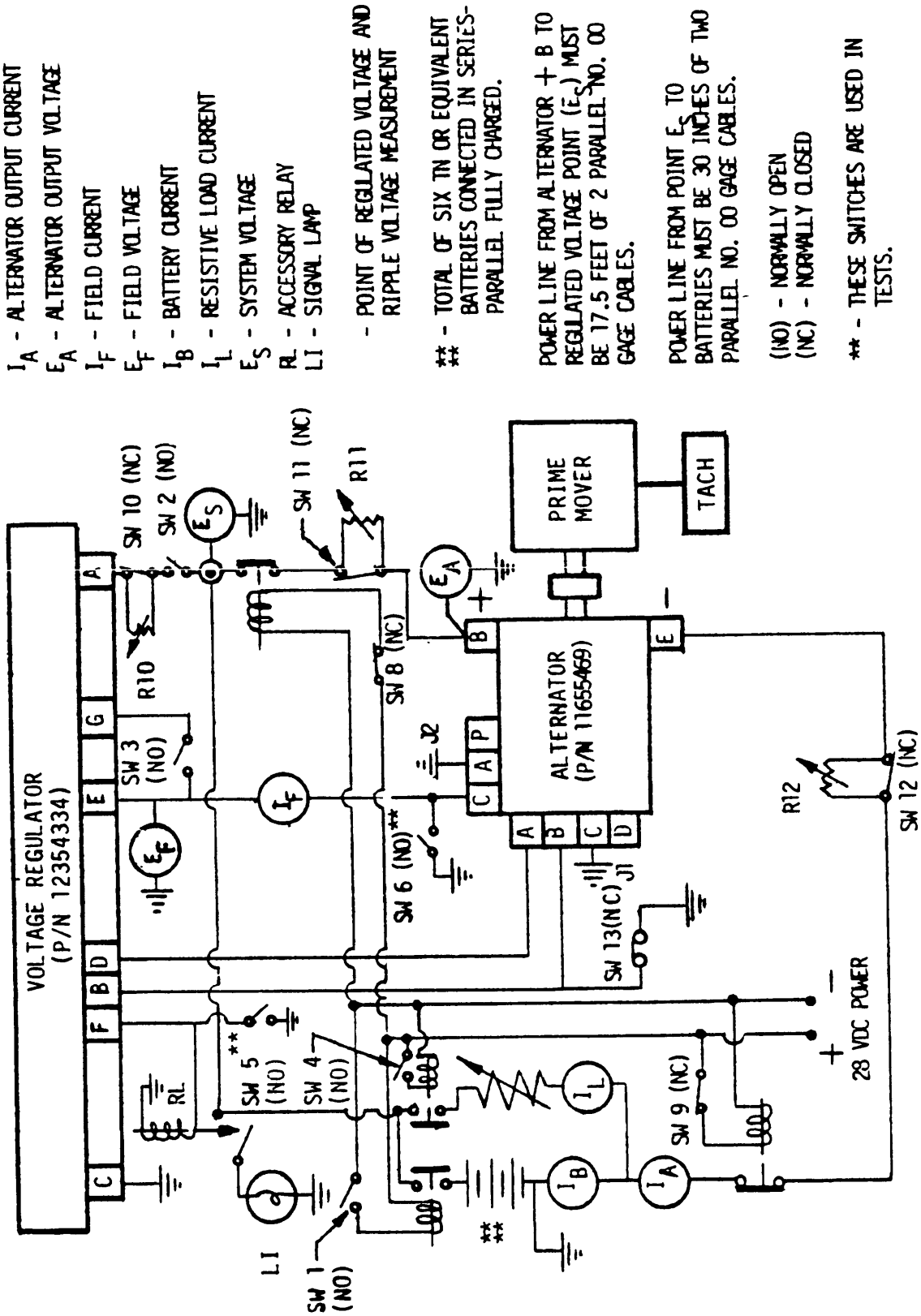


FIGURE 3. Voltage regulator test circuit.

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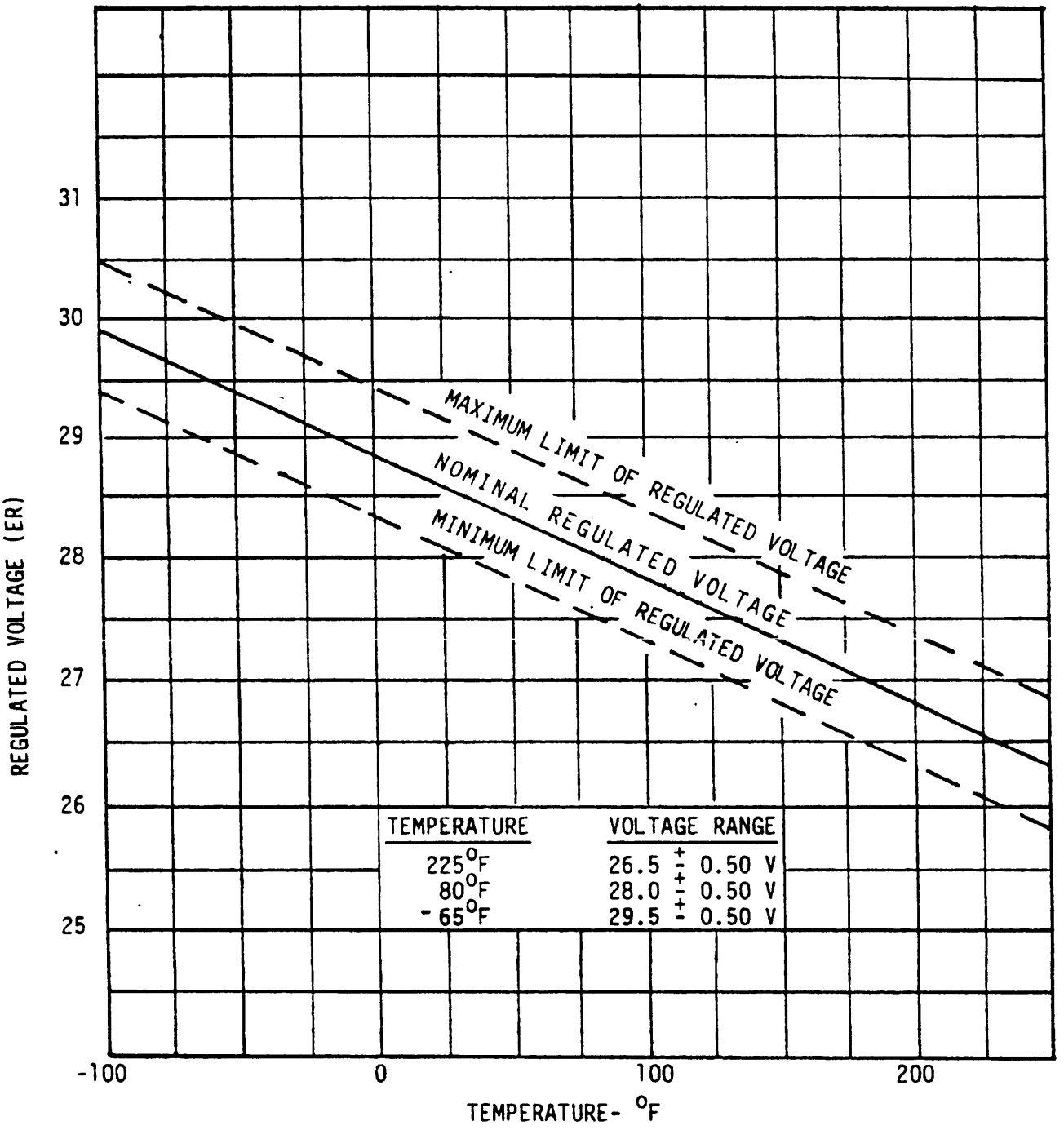


FIGURE 4. Voltage regulator temperature compensation (P/N 12354334).

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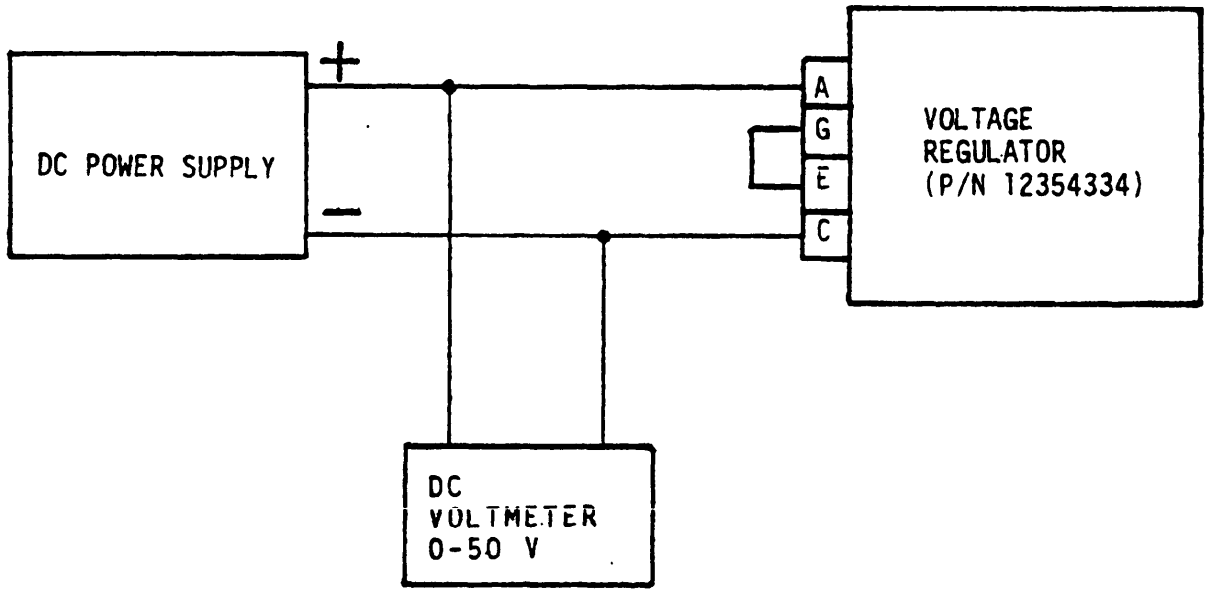


FIGURE 5. Overvoltage circuit test.

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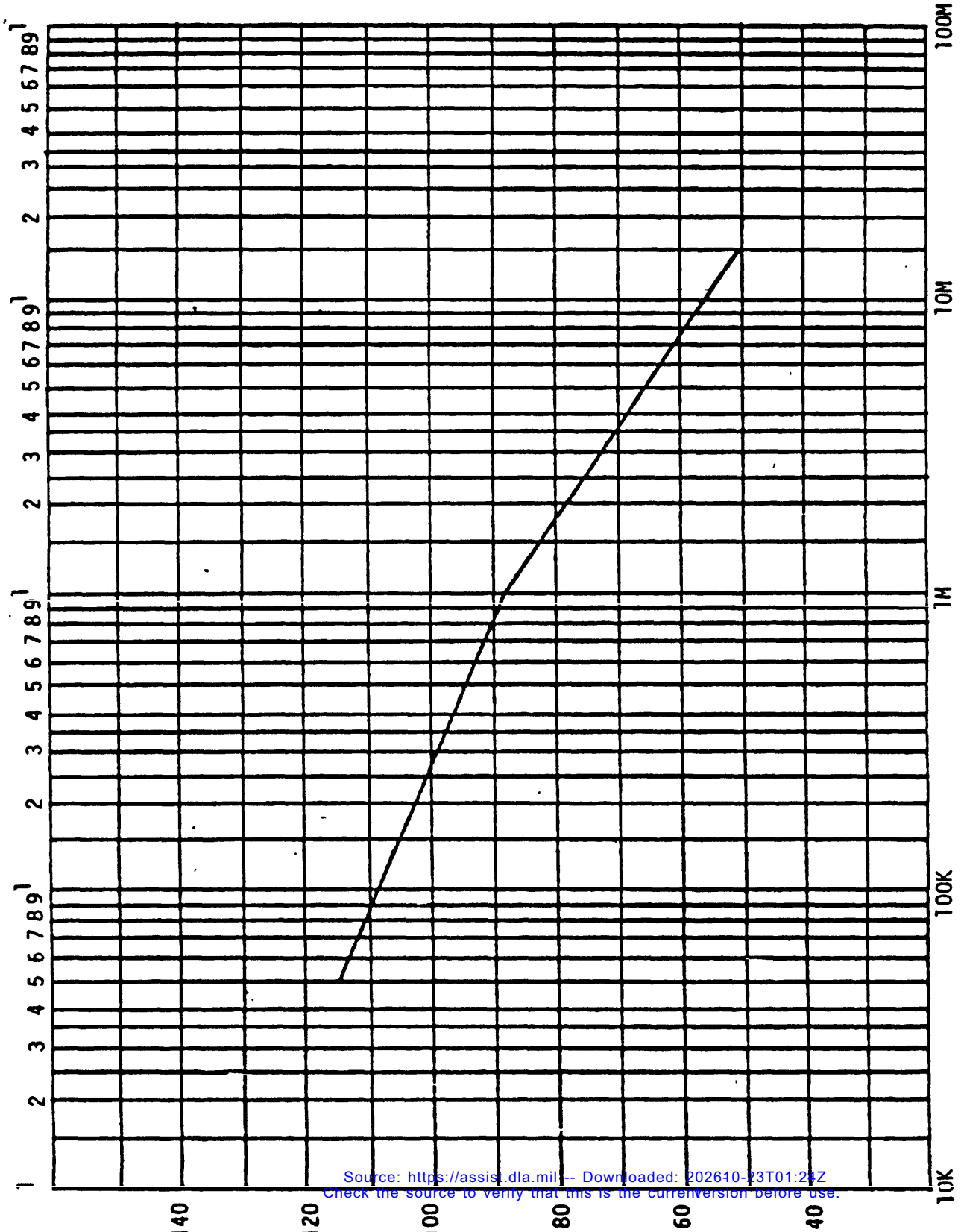


FIGURE 6. Broadband emission limits (conducted EMI).

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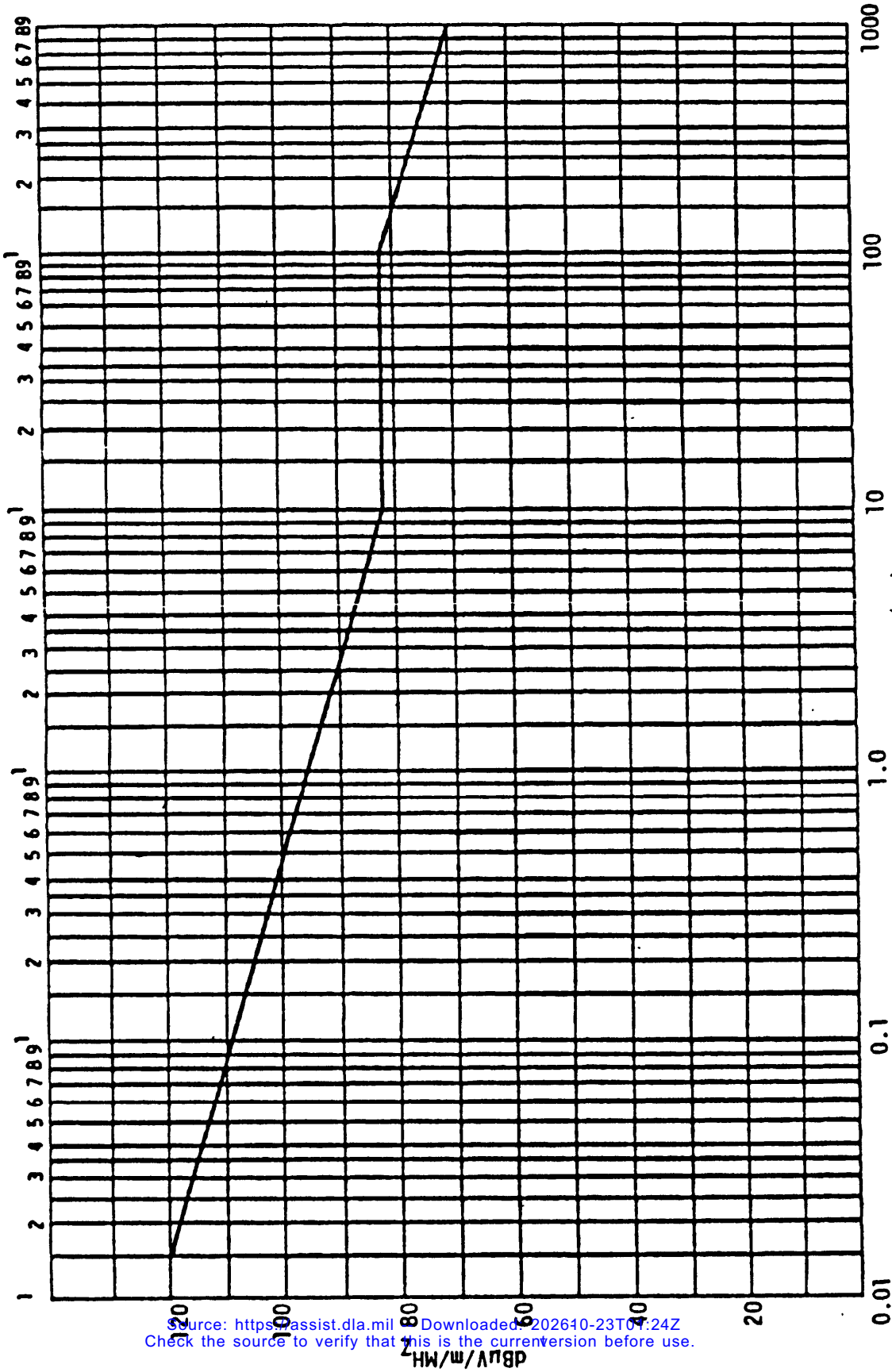
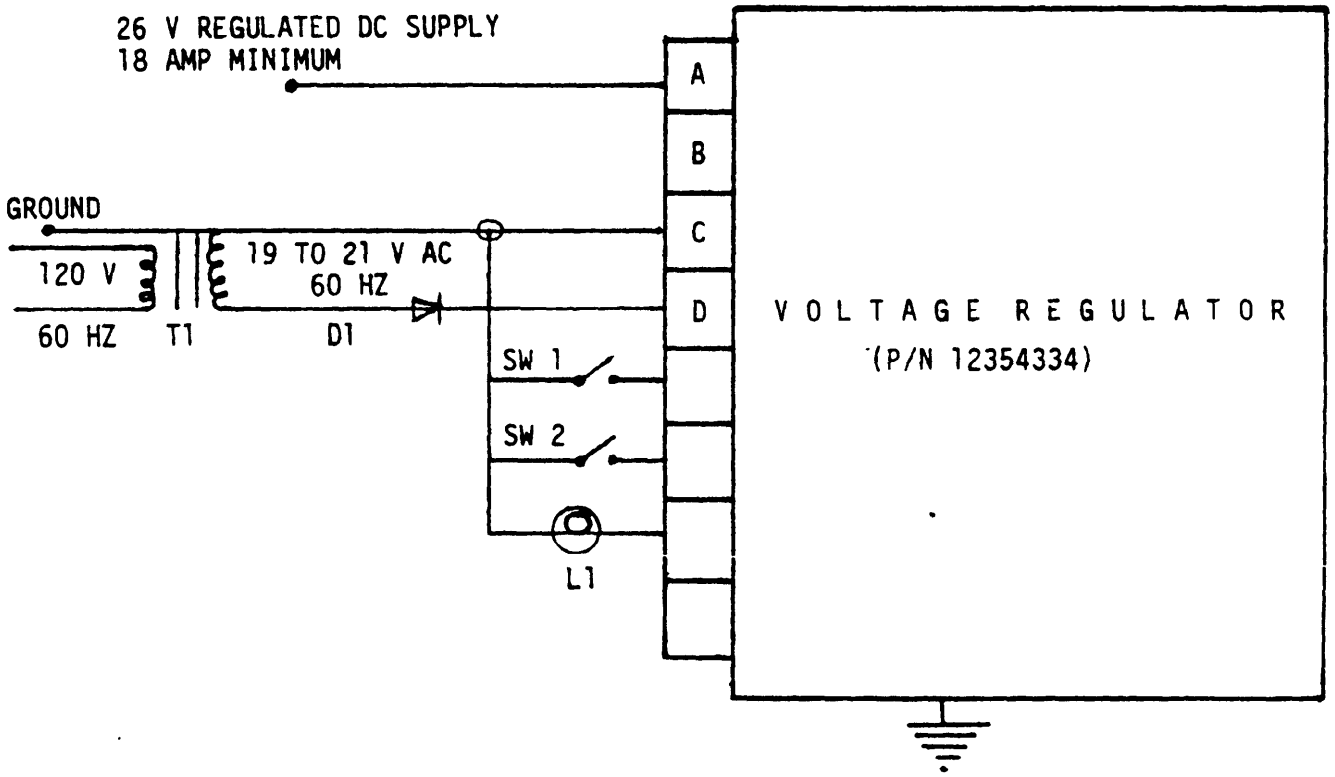


FIGURE 7. Broadband emission limits (radiated EMI).

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SW 1- NORMALLY OPEN SINGLE POLE SINGLE THROW (S.P.S.T.) MOMENTARY SW 18 AMP MINIMUM RATING.

SW 2- NORMALLY OPEN S.P.S.T. MOMENTARY SW 1 AMP MINIMUM RATING.

L1- 28 V DC LAMP- MS25237-327.

T1- TRANSFORMER 60 HZ 19 TO 21 V SEC, 1 AMP RATING.

D1- 3 AMP 50 VOLT DIODE.

FIGURE 8. Short circuit protection.

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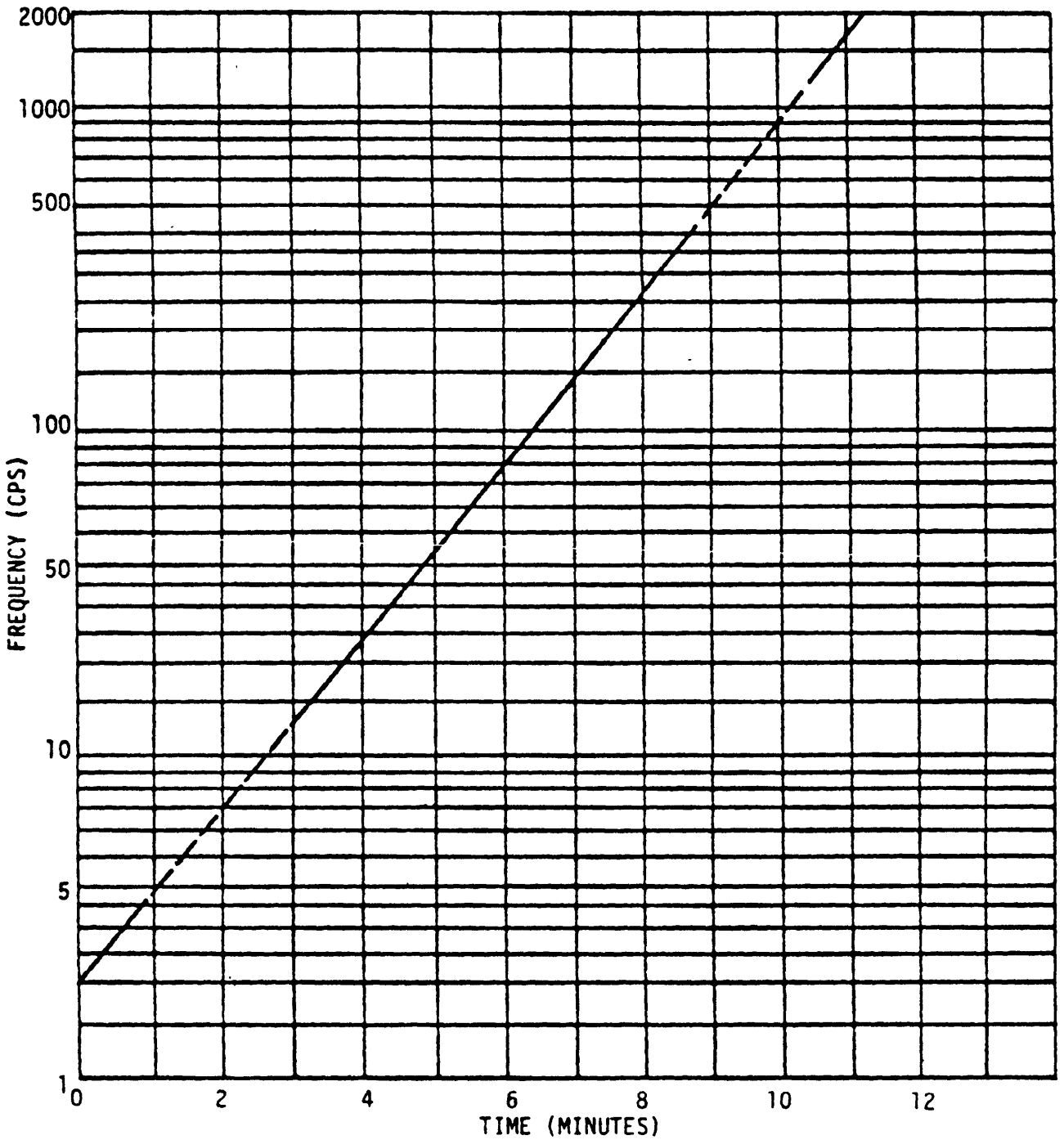
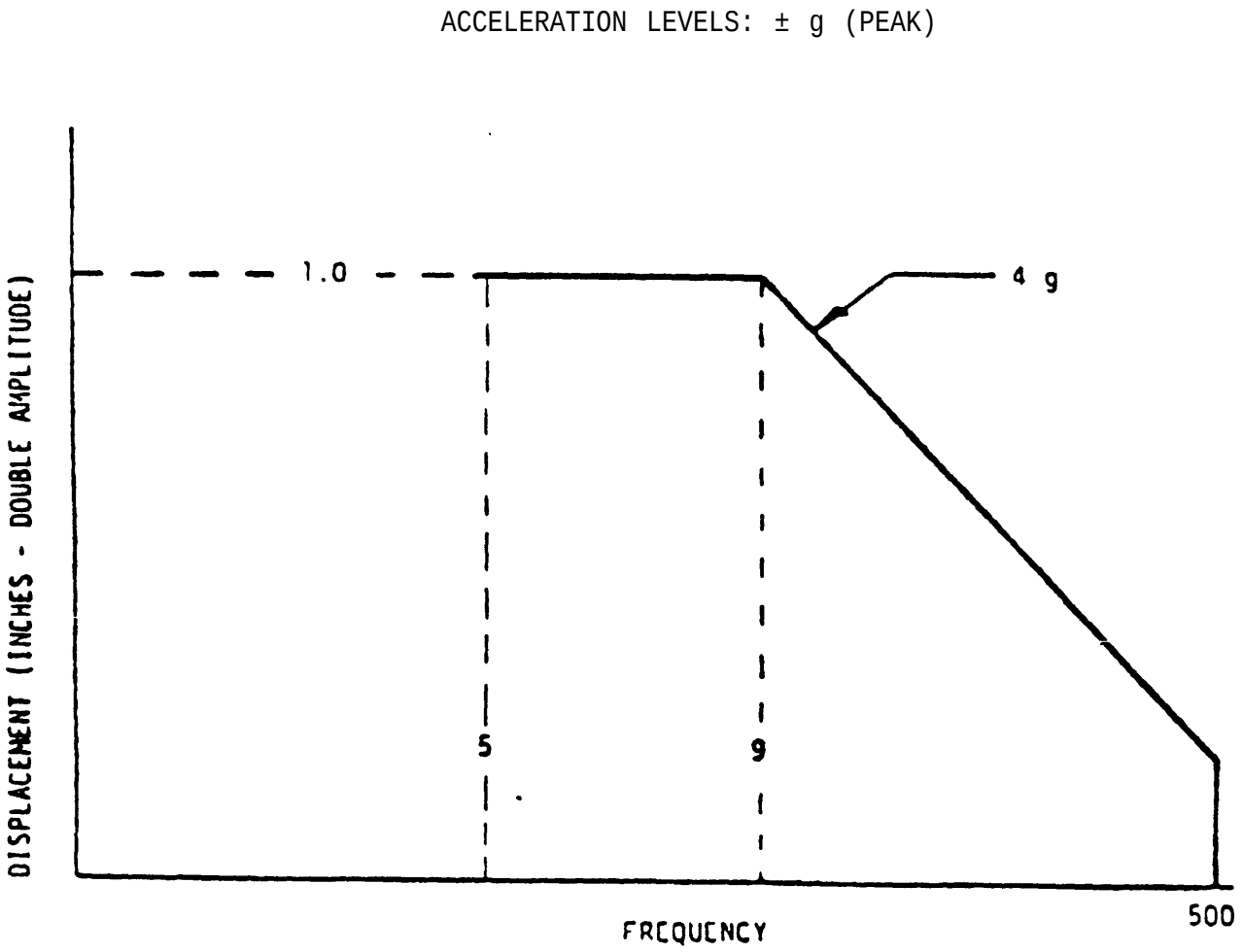


FIGURE 9. Logarithmic sweep.

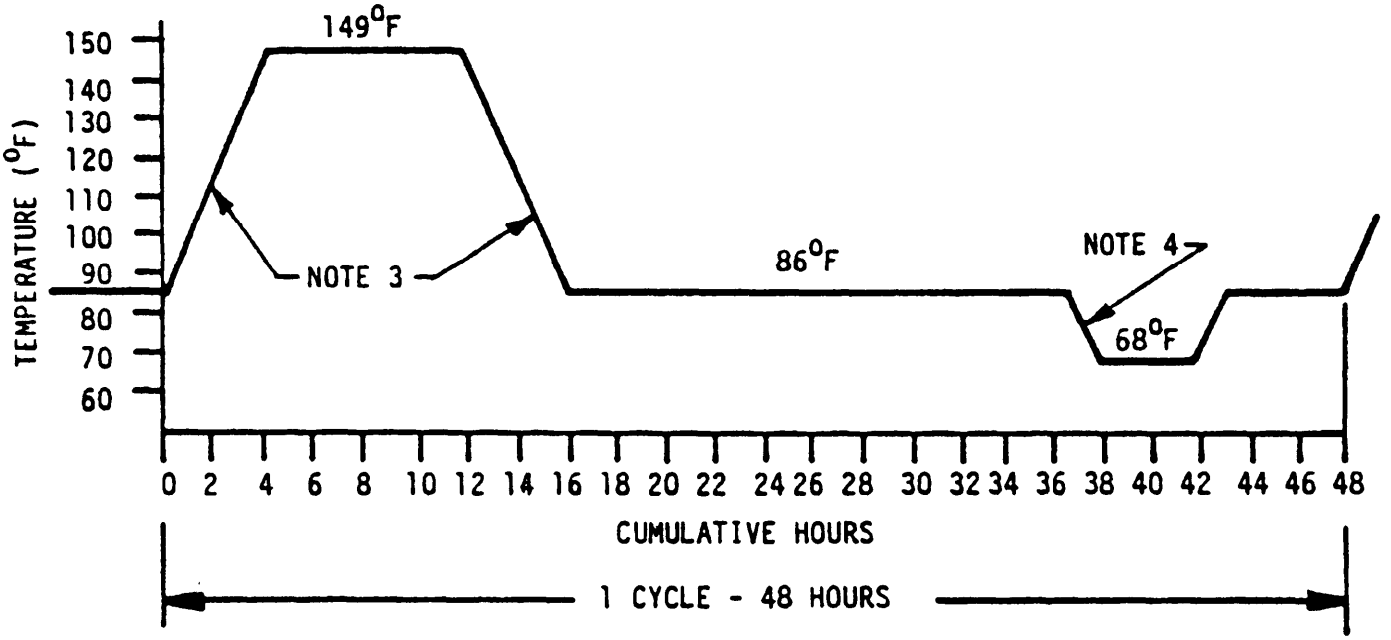
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NOTE: All curves shall be extended to 2 CPS when test item resonances below 5 CPS are expected.

FIGURE 10. Vibration test curve for equipment installed in ground vehicles.

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NOTES:

1. TOLERANCE DURING TEMPERATURE CHANGE SHALL NOT BE GREATER THAN 5°F.
2. RELETIVE HUMIDITY SHALL BE MAINTAINED AT 94 + 4% AT ALL TIMES, EXCEPT
3. THAT DURING THE DESCENDING TEMPERATURE PERIOD, THE RELETIVE HUMIDITY MAY BE PERMITTED TO DROP AS LOW AS 85%.
3. RATE OF TEMPERATURE CHANGE BETWEEN 86° AND 149°F SHALL NOT BE LESS THAN 14.4°F PER HOUR.
4. THE TEMPERATURE INCREASE IN THIS PORTION OF THE CURVE SHALL NOT BE LESS THAN 18°F.

FIGURE 11. Humidity cycle.

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Custodians:
Army - AT
Air Force - 99

Review activities:
Army-AR
Air Force - 82
DLA - CS

Preparing activity:
Army - AT

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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